Solution :

1. Inside the sphere $E = \frac{kQr}{R^3}$ and outside the sphere $E = \frac{kQ}{r^2}$.

4. Electric force is a central force, therefore torque w.r.t. centre is zero as a result angular momentum is conserved.

6. A positively charged conductor can have +ve, -ve or zero potential in the presence of any other nearly charge.

11. Potential at centre is $V_C = \frac{kQ}{R_A} - \frac{kQ}{R_B}$.

16. Considering potential to be zero at infinity. Work done by an external force in bringing a unit positive charge from infinity to a point without acceleration = – work done by electrostatic force.

17. The electrostatic potential (V) at any point in a region with electrostatic field is the work done in bringing a unit positive charge (without acceleration) from infinity to that point.

18. Since E = 0 inside the conductor and has no tangential component on the surface, no work is done in moving a small test charge within the conductor and on its surface.

19. For linear isotropic dielectric $p = \chi_e E$ (Direction of P & E are same)

20. In the absence of any external field, the different permanent dipole are oriented randomly due to thermal agitation, so the total dipole moment is zero.

21. The extent of polarisation depends on the relative strength of two mutually opposite factor. The potential energy in the external field tending to align the dipoles with the field and thermal energy tending to disrupt the alignment.

23. The material suitable for use as dielectric must have high dielectric strength x and high dielectric constant k.

24. $U = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{A\in_0}{d}\right)(Ed)^2 = \frac{1}{2}\in_0 E^2Ad$

$\left(c = \frac{A\in_0}{d}\right)(v = Ed)$

25. $\Delta V = \frac{q_1}{4\pi\in_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

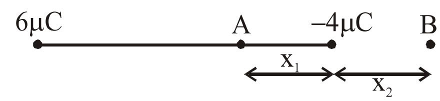
26. Van de graff generator is a machine that can build up high voltage of the order of a few million volts. The resulting large electric fields are used to accelerate charged particle, (electron, proton ions) to high energies needed for experiments to prob the small scale structure of matter.

27. Being a conductor, each plate have the same potential at each point. So potential gradient will be highest at closest end. As $E \propto \sigma$, so surface charge density σ is higher at the closer end.

28. $E \propto \frac{1}{r^2}$, $V \propto \frac{1}{r}$

So electric field decreases faster.

29.



For point A : $\frac{x_1}{15 - x_1} = \frac{4}{6} = \frac{2}{3} \Rightarrow x_1 = 6$ cm

For point B : $\frac{x_2}{15 + x_1} = \frac{2}{3} \Rightarrow x_2 = 30$ cm

So 45 cm from 6 μC

30. $W = (-q)(V_A - V_B)$ {B to A}

$= q(V_B - V_A) = +$ve as $(V_B > V_A)$

31. Total dipole moment $= 6 \times 10^{23} \times 10^{-29}$

$= 6 \times 10^{-6}$ C × m

$\Delta H = U_f - U_i$

$= -PE(\cos 60 - \cos 0)$

$= \frac{+PE}{2} = \frac{6\times10^{-6}\times10^5}{2} = 0.3$ J

33. Net electric field due to induced charges and the charge q is zero.

34. $C = \frac{A\in_0}{20\times10^{-3}} = 70\ \mu F$

$\Rightarrow A\in_0 = 1400 \times 10^{-2}$

$C' = \frac{A\in_0}{\frac{40-10}{1} + \frac{10}{2}} = \frac{A\in_0}{10^{-3}\times35} = \frac{1400}{35} = 40\ \mu F$

35. $C \rightarrow C_1 \quad C_2 \quad C_3$

$1 : 2 : 3$

$V \rightarrow \frac{1}{1} : \frac{1}{2} : \frac{1}{3}$

$\rightarrow 6 : 3 : 2$

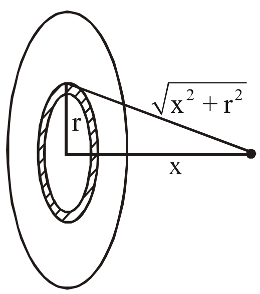
$V_{C_2} = \frac{3}{11} \times 66 = 18\ V$

$q_{C_2} = CV = 36\ \mu C$

36. $U = \frac{1}{2} C \Delta V^2$

$= \frac{4\pi \epsilon_0}{2} \frac{R \times 2R}{2R - R} \left(V - \frac{V}{2}\right)^2$

$= \pi \epsilon_0 R V^2$

37. 

$dV_p = \frac{k\,dq}{\sqrt{x^2 + r^2}}$

$V_p = \int_0^R \frac{k(\sigma\ 2\pi r dr)}{\sqrt{x^2 + r^2}}$

$V_p = \frac{2kQ}{R^2}\left(\sqrt{R^2 + x^2} - x\right)$

38. $C = \frac{4\pi \epsilon_0 r_1 r_2}{l n (r_2 / r_1)}$ and $V = \frac{Q}{c}$

39. $q_{sys} = \sigma \times 4\pi R^2 + \sigma \times 4\pi (2R)^2 = 20\sigma\pi R^2$

$q_1' = \frac{R}{R + 2R} \times 20\ \sigma\pi R^2 = \frac{20}{3} \sigma\pi R^2$

$q_2' = \frac{40}{3} \sigma\pi R^2$

$\sigma_1' = \frac{q_1'}{S_1} = \frac{20}{3} \frac{\sigma\pi R^2}{4\pi R^2} = \frac{5}{3}\sigma\ (S1 = 4\pi R^2)$

$\sigma_2 = \frac{5}{6}\sigma$